

***Intensity and timing of physical activity in relation to  
postmenopausal breast cancer risk: the prospective NIH-AARP  
Diet and Health Study***

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Background Despite strong evidence of an inverse association of physical activity with postmenopausal breast cancer risk, whether a certain intensity or time of life of physical activity is most effective for lowering breast cancer risk is not known. Methods In 118,899 postmenopausal women in the prospective NIH-AARP Diet and Health Study, we examined the relations of light and moderate-to-vigorous intensity physical activity during four periods of life ("historical": ages 15-18, 19-29, 35-39 years; "recent": past 10 years) to postmenopausal breast cancer risk. Physical activity was assessed by self-report at baseline, and 4287 incident breast cancers were identified over 6.6 years of follow-up. Results In age-adjusted and multivariate Cox regression models, >7 hours/week of moderate-to-vigorous activity during the past 10 years was associated with 16% reduced risk of postmenopausal breast cancer (RR:0.84; 95%CI:0.76,0.93) compared with inactivity. The association remained statistically significant after adjustment for BMI (RR:0.87; 95%CI:0.78,0.96). Neither moderate-to-vigorous activity during other periods of life nor light intensity activity during any period of life was related to breast cancer risk, and associations did not vary by tumor characteristics. Conclusion A high level of recent, but not historical, physical activity of moderate-to-vigorous intensity is associated with reduced postmenopausal breast cancer risk. More precise recall of recent physical activity than activity in the distant past is one possible explanation for our findings.